ZAALIMA DEVELOPMENT · DATA ANALYTICS INTERNSHIP

Week 2 • Day 7

SHAP Model Explainability

Generated SHAP values to explain every prediction — making the black box transparent for stakeholders

DATE	INTERN NAME	PROJECT	MENTOR
Day 14 of 28 | Week 2	____________________ _____	Customer Churn Prediction & LTV Engine	____________________ _____

■ Technologies Used Today

Python	SHAP	XGBoost	Pandas	Matplotlib	NumPy	joblib

■ Today's Key Metrics

3	1,409	14	4
SHAP Plots Created	Customers Explained	Total GitHub Commits	Top SHAP Features Found

■ Day Objectives

- ■ Understand SHAP values and why explainability matters for business
- ■ Calculate SHAP values for all test set customers
- ■ Create SHAP summary beeswarm and bar plots
- ■ Generate waterfall plot for the highest-risk customer
- ■ Save shap_explainer.pkl for API integration

■ Tasks Completed Today

1. Created src/models/shap_analysis.py
2. Loaded production_model.pkl and test data

3. Created shap.TreeExplainer(model) — optimised for XGBoost/tree models
4. Calculated SHAP values for all 1,409 test customers (~30-60 seconds)
5. Plot 1: SHAP Summary beeswarm — all features, all customers, with feature value colours
6. Plot 2: SHAP Summary bar — mean |SHAP| per feature as horizontal bars
7. Found highest-risk customer using np.argmax(y_prob)
8. Plot 3: SHAP Waterfall — individual customer explanation showing each feature's contribution
9. Confirmed top SHAP factors: tenure, contract_month-to-month, monthlycharges, techsupport
10. Saved reports/figures/14_shap_summary.png, 15_shap_bar.png, 16_shap_waterfall.png
11. Saved models/shap_explainer.pkl for API use
12. Documented SHAP business interpretations
13. Committed all files to GitHub — Week 2 complete

■ Concepts Learned

• SHAP: SHapley Additive exPlanations — mathematically rigorous feature attribution
• SHAP base value: the average prediction across the training set
• Positive SHAP value: this feature pushed this prediction TOWARD churn
• Negative SHAP value: this feature pushed this prediction AWAY FROM churn
• TreeExplainer: fast exact SHAP for tree models (XGBoost, RF)
• Beeswarm plot: each dot = one customer, colour = feature value, x = SHAP impact
• Waterfall plot: step-by-step from base value to final prediction for ONE customer
• Business translation: SHAP converts model output into specific retention actions

■ Tools / Software Used

Tool / Library	Version / Notes	Status
SHAP	0.44.0	✓ shap.TreeExplainer used

■ Outputs / Deliverables Produced

❖ src/models/shap_analysis.py — SHAP analysis script
❖ models/shap_explainer.pkl — saved SHAP TreeExplainer
❖ reports/figures/14_shap_summary.png — beeswarm plot
❖ reports/figures/15_shap_bar.png — mean SHAP bar chart
❖ reports/figures/16_shap_waterfall_highrisk.png — individual explanation
❖ Business interpretation documented for top-5 SHAP features
❖ GitHub commit — Week 2 complete, 14 total commits

■ Problems Encountered & Solutions

Problem Faced	How I Solved It

SHAP calculation very slow on large dataset	Used only X_test (1,409 rows) instead of full dataset for SHAP — still representative and much faster

■ GitHub Commit Record

Commit Message	Week2-Day7: SHAP explainability — summary, bar, waterfall plots, explainer saved, Week 2 complete
Branch	main
Files Changed	src/models/shap_analysis.py, models/shap_explainer.pkl, reports/figures/14-16 SHAP plots
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ shap_analysis.py created	■ SHAP values calculated
■ Beeswarm summary plot done	■ Bar chart (mean SHAP) done
■ High-risk customer found	■ Waterfall plot created
■ shap_explainer.pkl saved	■ Charts 14-16 saved
■ SHAP insights documented	■ Week 2 complete — 14 commits

■ Self Assessment

10/10	**6**	**Week 2, Day 7**	**14/28**
Self Rating	Hours Spent	Progress	Day in Program

■ Key Learnings & Reflections

★ SHAP transforms ML from a black box to a business tool with actionable explanations

★ tenure with low value has the highest positive SHAP for churn — short-tenure customers are highest risk

★ Contract type month-to-month shows strong positive SHAP — target these for annual contract offers

★ SHAP must be saved as an explainer object — not just the values — so the API can recalculate for new customers

■ Plan for Tomorrow

→ Week 3 Day 1: Calculate Customer Lifetime Value (LTV) for all 7,043 customers
→ Apply churn probabilities to build predictive LTV formula
→ Segment customers into High/Medium/Low Value tiers
→ Save customers_ltv to PostgreSQL

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED
____________________	____________________	____________________